solution of sum of square error under regularization

$$E_D(W) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n,w)-t_n\}^2 + \frac{\lambda}{2}\|w\|^2$$

1. partial derivate with respect to w_i

$$\frac{\partial}{\partial w_i}E_D(w) = \frac{1}{2}\sum_{n=1}^{N} 2\{y(x_n,w)-t_n\}\cdot\underbrace{\frac{\partial}{\partial w_i}y(x_n,w)} + \lambda w_i$$

$$\frac{\partial}{\partial w_i} y(x_n,w) = \frac{\partial}{\partial w_i}(w_0 + w_1 x_n^1 + w_2 x_n^2 \cdots\cdots w_i x_n^i \cdots + w_{M-1}x_n^{M-1})$$

$$= x_n^i$$

2. set $\frac{\partial}{\partial w_i}E_D(w) = 0$

$$\frac{\partial}{\partial w_i}E_D(w) = 0 \Rightarrow \sum_{n=1}^{N}\{y(x_n,w)-t_n\}(x_n)^i + \lambda w_i = 0$$

$$\Leftrightarrow \sum_{n=1}^{N}(x_n)^i \cdot y(x_n,w) + \lambda w_i = \sum_{n=1}^{N} t_n (x_n)^i$$

3. let $\sum_{n=1}^{N}(x_n)^{(i+j)} = a_{ij}$, $\sum_{n=1}^{N} t_n (x_n)^i = T_i$

$$\sum_{n=1}^{N}(x_n)^i y(x_n,w) = \sum_{n=1}^{N}(x_n)^i \cdot \sum_{j=0}^{M-1} w_j (x_n)^j$$

$$\Rightarrow \sum_{j=0}^{M-1}\sum_{n=1}^{N} w_j (x_n)^{i+j} = \sum_{j=0}^{M-1} w_j a_{ij}$$

$$\therefore \sum_{j=0}^{M-1} w_j a_{ij} + w_i = T_i$$

4. 对 $i = 0, 1, 2, \cdots M-1$, $w = [w_0, w_1, \cdots w_{M-1}]^T$

$(i=0)$

$$\sum_{j=0}^{M-1} w_j a_{0j} + \lambda w_0 = T_0$$

$(i=1)$

$$\sum_{j=0}^{M-1} w_j a_{1j} + \lambda w_1 = T_1$$

$\vdots$

$(i=M-1)$

$$\sum_{j=0}^{M-1} w_j a_{M-1j} + \lambda w_{M-1} = T_{M-1}$$

let $T = [T_0, T_1, \cdots T_{M-1}]^T$

$A_i = [a_{0j}, a_{1j}, \cdots, a_{M-1j}]$

let $A_{ij} = \sum_{n=1}^{N} (x_n)^{(i+j)}$

$$\Rightarrow A_0 w + \lambda w_0 = T_0 \Rightarrow \underline{(A + \lambda I) w = T}$$

$A_1 w + \lambda w_1 = T_1$ where $A_{ij} = \sum_{n=1}^{N} (x_n)^{i+j}$

$A_2 w + \lambda w_2 = T_1$ $T_i = \sum_{n=1}^{N} t_n (x_n)^i$

$\vdots$

$A_{M-1} w + \lambda w_{M-1} = T_{M-1}$